COMPREHENSIVE NUTRITION COURSE

Childhood Nutrition

Obesity & Short Stature

PRESENTED BY

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CHILDHOOD OBESITY

Obesity is State of Adiposity in which body fat rises above ideal BMI ..

Over weight is a state in which weight exceeds Standards , based on height , BMI

EXCEEDING

20%

Obese Child : is child with body weight exceeding 20% of normal average wt of his age

NORMAL AVERAGE WEIGHT

is calculated by one of the following methods :

FOR CHILD 1 - 5 YEAR

weight (kg) =

(age in year + 5) **X** 2

FOR CHILD 5 - 12 YEAR

weight (kg) =

age in year **X** 4

CLASSIFICATION · I

EXOGENOUS : ENERGY IMBALANCE

01

High caloric density food

02

Supersized Portions

03

Eating out (junk food)

04

Advertising

05

Working Parents

06

TV

07

Computers

08

Physical inactivity

09

soft drinks – Candy desserts high in calories

10

Medications

CLASSIFICATION · II

ENDOGENOUS

GENETICS

80%

of children with 2 over weight parents will be obese

40%

of children with 1 over weight parents will be obese

II

Endocrinal

→ **Hypothyroidism**

→ **Cushing syndrome**

(Notice : Moon face – buffalo hump)

HAZARDS OF OBESITY

01 DM type II & elevated insulin level

02 HTN – dyslipidemia : ↑ LDL , TGs & Heart Disease

03 Neurological Complications .

04 Respiratory Disease (asthma)

05 Orthopedic Condition .

06 Psychological Disorders (Depression – anxiety – negative body image · Withdrawal from interactions – academic deterioration)

07 Skin : AN.

08 GI manifestations (fatty liver – elevated liver enzyme – Gall stone – Reflux)

09 menstrual Disorders & PCO

10 sleep apnea

HISTORY

1



Number of meals intake

2



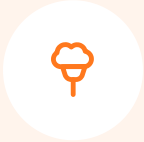
Snacks ○ yes ○ no

3



if yes → what kind of snack ?

4



Frequency of green vegetable intake

5



Frequency of fried food

6



eating sweets



ASSESSMENT OF OBESITY

Weight

Height

BMI ..

Clinical Examination : (Pediatrician)

LAB ASSESSMENT

1 CBC

2 Thyroid

3 Fasting , 2h PP glucose

4 Total Cholesterol

5 ALT & AST



DAILY CALORIC REQUIREMENT

carbohydrate	45-65%
Fat	25-35%
Proteins	10-20%



NUTRITION GUIDANCE

- encourage water intake
- limit sweet beverages
  (No sweets needed for any juice < 1 year)
- **Eat 5 fruits and vegetables per day**
- Know what the Child eat outside the home
- encourage child's self regulation of food intake (Choices)



ACTIVITY GUIDANCE

encourage active play for young children ..

decrease sedentary activity .

LIMIT TV , VIDEOGAMES & COMPUTER

1-2 h \ day

Avoid TV in rooms.

STAGES OF WEIGHT MANAGEMENT

STAGE	STAGE I	STAGE II	STAGE III
Goals	• weight maintenance • Decreasing BMI	• Gradual weight loss • Decreasing BMI	• Rapid Wt Loss
Strategy	• Monthly follow up	• weight loss shouldn't Exceed $\frac{1}{2}$ Kg \ month (2-5 years) & not Exceeds $\frac{1}{2}$ kg \ week In older Child • Monthly assess wt state → if no improvement in 3-6 Month ; move to stage III	

STAGES OF WEIGHT MANAGEMENT · METHOD

STAGE I

- Consume 5 serving fruit & vegetables \ day
- eliminate sugary Beverages
- Limit screen time → 2 h With no TV at Child Room
- 1 H daily Exercise
- eating daily breakfast
- eating family meal at Least 5-6 times \ week
- allow Child to regulate His \ her meal

STAGE II

- Balanced macronutrient Diet
- eat 2 snacks \ day
- Limit screen time → 1 h
- 1 H daily Exercise
- Reinforcement for achieving Targeted behavioral goals

STAGE III

- Severe Caloric Restriction
- Drug Therapy
- Surgery



CHILDREN AND SHORT STATURE

Short stature is **Familial** & **mal nutrition**

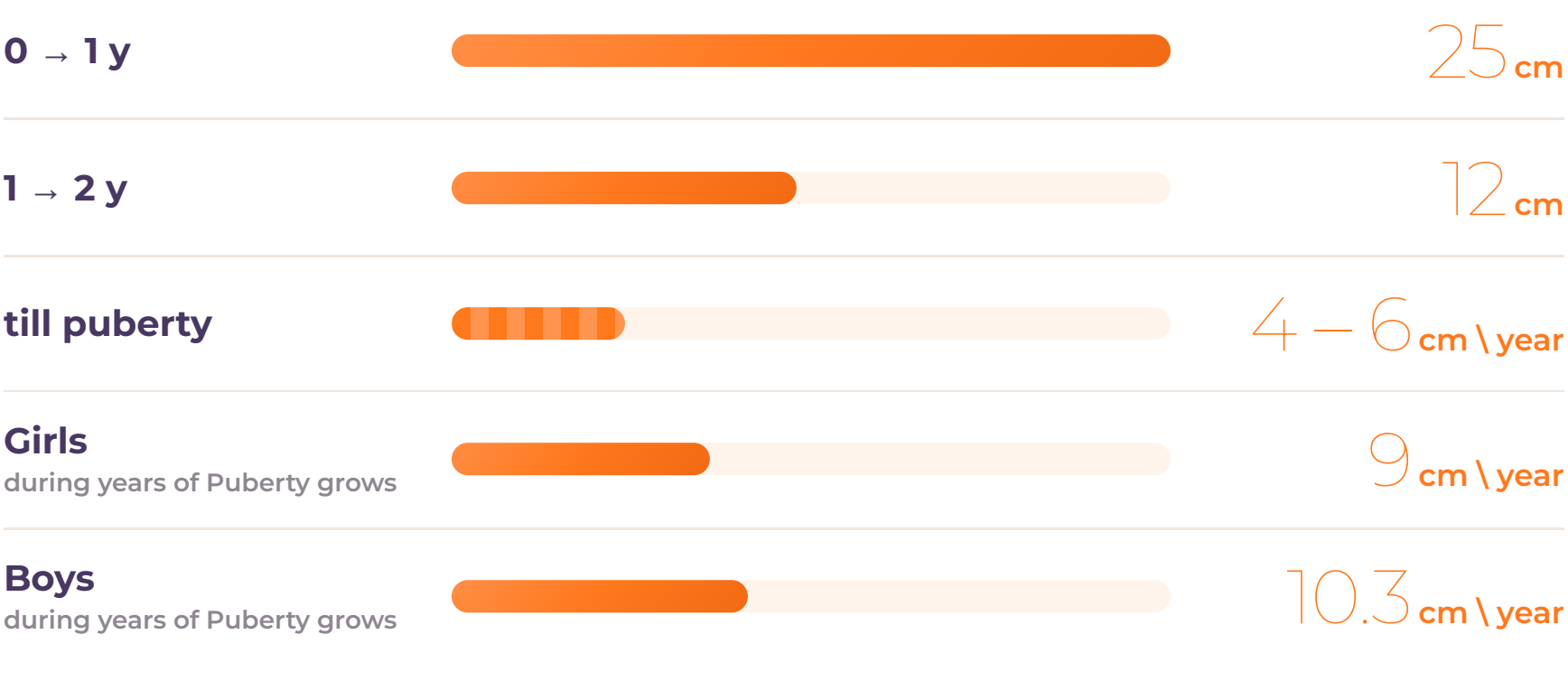
20% **of Egyptian children are short .**

MAY BE ASSOCIATED WITH :

- Low immunity
- educational problems
- easy fatigue
- anaemia
- Obesity
- Psychological upset

RATE OF GROWTH

(46 – 54 Cm)



NUTRIENT NEEDS



1 CHO

is the main source for energy ..

EXAMPLES

- Rice
- cereals
- macaroni
- bread
- potatoes



2 Fat

Olive oil – sunflower oil ,

Which help in absorption of fat soluble vitamins : **A , D , E , K**

Deficiency of these sources → breakdown



3 Proteins

essential for muscle & bone growth

SOURCE

- meat
- fish
- egg
- dairy product
- legumes
- lentils

4 MINERALS & VITAMINS IMPORTANT FOR GROWTH

	Ca Calcium	Zn Zinc	I iodine	Mg magnesium	A Vit. A	E Vit. E	D Vit. D
Source							
Sign of Deficiency							

إرشادات

TIPS

1. تناول الأطعمة التي تحتوي على فيتامين ج يساعد على امتصاص الكالسيوم والحديد
2. تناول الخميرة البيرة يوميا مع ملعقة عسل نحل مهم جدا للنمو
3. تجنب تناول المشروبات الغازية لأنها تمنع امتصاص الكالسيوم
4. تجنب تناول الأغذية الملونة المحتوية على مواد صناعية
5. أضف الملح المدعم باليود لأكلك وتناول السمك أو التونة مرتين أسبوعيا
6. اللبن ليس المصدر الوحيد للكالسيوم .. هناك الطحينة - السمسم - البيض

Thank You

LIFESTYLE FOUNDATION

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